NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND
MIL-PRF-39018/3D
AMENDMENT 2
7 January 1991
SUPERSEDING
AMENDMENT 1
25 March 1981

PERFORMANCE SPECIFICATION SHEET

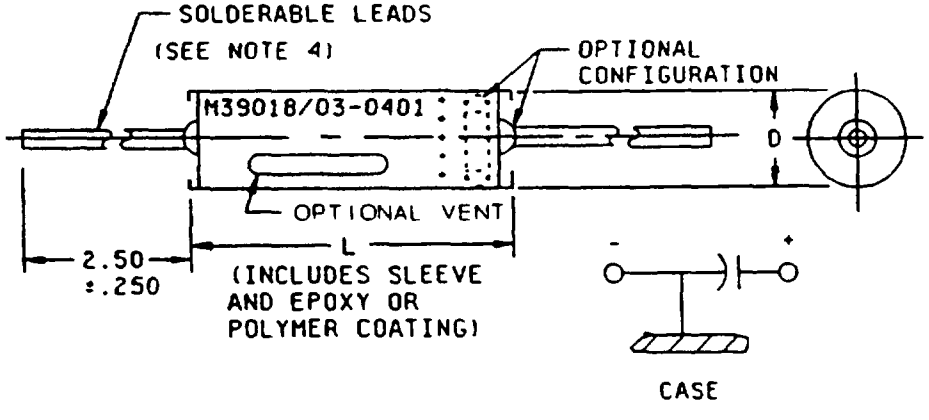
CAPACITORS, FIXED, ELECTROLYTIC (ALUMINUM OXIDE)
(POLARIZED), NON-ESTABLISHED RELIABILITY AND ESTABLISHED RELIABILITY,
STYLES CU16 (UNINSULATED), CU17 (INSULATED), AND CUR17 (INSULATED)

This amendment forms a part of MIL-C-39018/3D, dated 23 March 1978, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

*Figure 1, delete and substitute:

"



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These capacitors are not intended to be mounted by their leads.
4. For .625 diameter case, the leads shall be .032 +.008, -.002 diameter and for all other diameter cases the leads shall be .040 +.008 -.002 diameter.
5. The "D" dimension for styles CU17 and CUR17 shall be not more than .008 (0.20 mm) larger than the specified bare case diameter.

FIGURE 1. Styles CU16, CU17, and CUR17 capacitors."

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PAGE 2

Metric chart, add the following dimensions:

Inches	mm
" .250	6.35
2.50	63.5 "

PAGE 3

Stability at reduced and high temperatures, step 2: Delete "50 to 350 V dc inclusive: ±20 percent" and substitute:

"50 to 100 V dc inclusive: ±20 percent.
150 to 350 V dc inclusive: ±30 percent."

PAGE 6

* Table I, footnote 3: Add "For style CUR17 parts with random vibration, the letter R shall be used in place of the dash (i.e., M39018/03R1052M or M39018/03R1252M).".

PAGE 8

* Following vibration, high frequency: Add "Random (style CUR17 only) (when applicable) - in accordance with MIL-C-39018.".

* Thermal shock and immersion, thermal shock, following condition A: Add "except in step 3, units shall be tested at applicable high temperature".

PAGE 9

* Marking, part number: Add "For CUR17 parts with random vibration, the letter R shall be used in place of the dash (i.e., M39018/03R1052M or M39018/03R1252M).".

PAGE 10

* Add the following application note:

"APPLICATION NOTE:

Ripple current: The maximum permissible ripple current for these capacitors can be determined from table III. The appropriate multiplier from this table should be used where operation at other frequencies or temperatures is desired. Operation of these capacitors outside the limits of 60 to 100,000 Hz is not recommended.

TABLE III. Ripple current multipliers.

With temperature	+25°C	+45°C	+65°C	+85°C	---
	1.4	1.4	1.2	1.0	---
With frequency		60 Hz	120 Hz	400 Hz	1 K through 100 K Hz
	0-60 volts	.85	1.0	1.1	1.15
	61-200	.83	1.0	1.15	1.2
	201-350	.80	1.0	1.30	1.4

"

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV
Navy - SH
Air Force - 17, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5910-1688-3)

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